

Worksheet · Section 3.5–3.7

Booleans, Conditionals & Nested Conditionals



The Big Picture

Programs make decisions with Boolean expressions. Practice evaluating AND/OR/NOT, tracing IF/ELSE and nested IFs, and writing your own conditions.

Part 1 · Vocab Match

Write the letter of the matching definition on each line.

- | | |
|--|---|
| ____ 1. Boolean value | A. Reverses a Boolean value |
| ____ 2. Relational operator | B. Compares two values: =, ≠, <, >, ≤, ≥ |
| ____ 3. Logical operator | C. True when at least one condition is true |
| ____ 4. AND | D. True only when both conditions are true |
| ____ 5. OR | E. Expressions that give the same result for every input |
| ____ 6. NOT | F. A conditional inside another conditional |
| ____ 7. Selection | G. Choosing which code runs based on a condition |
| ____ 8. Conditional statement | H. An IF or IF/ELSE that controls which code runs |
| ____ 9. Nested conditional | I. NOT, AND, or OR |
| ____ 10. Equivalent expressions | J. A value that is either true or false |
| ____ 11. Branch | K. One possible path through a conditional |

Part 2 · Fill It In

Use each word from the word bank once.

true

=

OR

false

nested

ELSE

NOT

AND

- Boolean values are true or _____.
- _____ is true only when both conditions are true.
- _____ is true when at least one condition is true.
- _____ flips true to false.
- In AP pseudocode, _____ checks whether two values are equal.
- The _____ block runs when the IF condition is false.
- An IF inside another IF is a _____ conditional.
- $(5 > 3)$ OR $(2 > 9)$ evaluates to _____.

Name: _____

Class: _____

Date: _____

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Practice

Part 3 · Evaluate

$x \leftarrow 7$ and $y \leftarrow 12$

Expression	Value
$x > 5$ AND $y < 10$	_____
$x > 5$ OR $y < 10$	_____
NOT ($x = 7$)	_____
$(y \text{ MOD } 2 = 0)$ AND $(x \text{ MOD } 2 = 1)$	_____
NOT ($x > y$ OR $y > 20$)	_____

Part 4 · Truth Table

A	B	NOT B	A AND NOT B	A OR B
T	T	_____	_____	_____
T	F	_____	_____	_____
F	T	_____	_____	_____
F	F	_____	_____	_____

Part 5 · Trace the Branches

```

IF (score ≥ 90)
{ result ← "A" }
ELSE
{
  IF (score ≥ 75)
  {
    IF (bonus) { result ← "A" }
    ELSE { result ← "B" }
  }
  ELSE { result ← "C" }
}

```

score	bonus	result
95	false	_____
80	true	_____
80	false	_____
60	true	_____
75	false	_____

Part 6 · Write the Condition

a) n is between 1 and 50, inclusive:

b) day is a weekend ("Sat" or "Sun"):

c) Rewrite without NOT: NOT ($a > 3$ AND $b < 2$)

d) temp is NOT between 60 and 80:



Think:

Trace every branch: for each input, follow ONE path through the IFs. Separate IF statements can each run; an IF/ELSE runs exactly one block.

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Exam Practice

Part 7 · Exam Practice

★ Warm-up

1. What does `NOT (4 = 4)` evaluate to?

- A. false
- B. Error
- C. true
- D. 4

★★ On Level

2. What is `x` after this code?

```
x ← 10
IF (x > 5) { x ← x - 3 }
IF (x > 5) { x ← x - 3 }
```

- A. 7
- B. 10
- C. 4
- D. 1

★★ On Level

3. Which expression is true when `age` is NOT between 13 and 19, inclusive?

- A. `age < 13 AND age > 19`
- B. `age ≥ 13 OR age ≤ 19`
- C. `age < 13 OR age > 19`
- D. `age ≥ 13 AND age ≤ 19`

★★★ Exam Hard

4. For which input is "Y" displayed?

```
IF (a > b)
{
  IF (a > 10) { DISPLAY("X") }
  ELSE { DISPLAY("Y") }
}
ELSE { DISPLAY("Z") }
```

- A. `a = 8, b = 8`
- B. `a = 12, b = 3`
- C. `a = 8, b = 3`
- D. `a = 3, b = 8`

★★★ Exam Hard

5. Which TWO are equivalent to `IF (a) { IF (b) { DISPLAY("ok") } }`? (Select two.)

- A. `IF (NOT (NOT a OR NOT b)) { DISPLAY("ok") }`
- B. `IF (a AND b) { DISPLAY("ok") }`
- C. `IF (a OR b) { DISPLAY("ok") }`
- D. `IF (b) { DISPLAY("ok") }`

Part 8 · Written Response

A gym charges members \$10. Non-members pay \$15, but non-member students pay \$12. Using `isMember` and `isStudent`, write nested conditionals that set `price`.
